Following the Sun: What Bangladesh Can Learn from Neighbouring Pakistan's Solar Boom

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Abstract

Both Bangladesh and Pakistan have endured severe electricity crises, yet of fundamentally opposite types. Pakistan's was a grid-and-price crisis: soaring tariffs and a failing, debt-ridden utility drove households and factories to abandon the national grid for cheap rooftop solar, producing one of the world's fastest distributed-solar booms. Bangladesh's is a fiscal-and-fuel paradox: it holds substantial idle generation capacity, yet suffers recurring load shedding because it lacks the foreign exchange to buy the imported fuel that capacity needs, a vulnerability the 2026 Iran–United States war and Strait of Hormuz gas-supply disruption laid bare. This paper asks what a country with the second kind of crisis can learn from a neighbour that stumbled into solar through the first. Using a mixed-methods comparative case study and a momentum-versus-management framework, it argues that Pakistan offers two lessons: positively, self-generated solar quickly eased grid pressure and cut imported-fuel reliance, strengthening energy security; as a caution, an unmanaged boom eroded the utility's revenue base, pushed wealthier users off the grid, and triggered abrupt policy reversals. The paper contends that Bangladesh should treat rooftop solar not as a climate amenity but as an energy-security and macro-financial imperative, pursued through a managed, industrially-anchored pathway that begins with the export garment sector and then scales through public and commercial buildings. It closes with engineering, financing, and governance recommendations for capturing Pakistan's gains while avoiding its mistakes.

Keywords: *rooftop solar; energy security; distributed generation; Bangladesh garment sector; managed energy transition*

1. Introduction

In early March 2026, as a war between Iran on one side and the United States and Israel on the other closed the Strait of Hormuz to commercial shipping, Bangladesh slid into an energy emergency. QatarEnergy, one of the country's three long-term liquefied natural gas (LNG) suppliers, declared force majeure after attacks on its facilities, and within days the other two also stopped deliveries (Global Voices 2026). Bangladesh, which had scheduled 115 LNG cargoes for 2026, projected losing forty; gas supply to power plants fell well below demand, four of five state urea factories shut, and officials warned of a likely 1,800-megawatt (MW) shortfall and rolling load shedding (Global Voices 2026; The Business Standard 2026a). Neighbouring Pakistan, by contrast, faced the same global fuel shock from a position of unusual insulation: having quietly installed tens of gigawatts of rooftop solar over the preceding

five years, it had already cut its grid demand and its appetite for imported gas (World Economic Forum 2025).

Two neighbouring South Asian economies, the same falling panel prices, the same 2026 fuel shock — yet sharply different outcomes. This contrast frames the paper's central question: what lesson does Pakistan's boom offer a country whose crisis is driven not by tariffs but by import dependence? The two countries suffered crises of opposite types: Pakistan's grid-and-price crisis pushed consumers to flee an expensive, unreliable utility, while Bangladesh's fiscal-and-fuel paradox leaves it owning more plants than it can afford to run. Bangladesh therefore cannot simply copy Pakistan's consumer-led rush but it can learn from both its gains and its governance failures.

This study addresses three research questions. **RQ1:** What structural conditions drove Pakistan's rapid distributed-solar adoption, and what were its system-level consequences? **RQ2:** How does Bangladesh's situation differ, and why is rooftop solar an energy-security rather than merely a climate question for it? **RQ3:** Which elements of Pakistan's experience are transferable, which are cautionary, and what managed, sector-anchored pathway follows? The contributions are: (i) a comparative reframing of rooftop solar as an energy-security and climate-resilience response to import-fuel shocks; (ii) a momentum-versus-management analytical framework; (iii) a phased, garment-anchored deployment roadmap combining engineering, techno-economic, and governance detail; and (iv)A transferable-lessons typology distinguishing what to replicate from what to avoid.

2. Related Work

Pakistan's solar surge has attracted substantial recent analysis. Institutional studies document its scale and drivers, attributing it to a "perfect storm" of collapsing panel prices, rising grid tariffs and an unreliable supply (World Resources Institute 2025; REN21 2025). A parallel literature examines the boom's system-level strains — cross-subsidy shifts, grid-revenue erosion and the difficulty of even measuring distributed capacity (TransitionZero 2025; World Economic Forum 2025). On the Bangladesh side, analyses of rooftop solar focus on policy design, the 2025 National Rooftop Solar Program, and persistent barriers in financing and net metering (Institute for Energy Economics and Financial Analysis [IEEFA] 2025; Centre for Policy Dialogue [CPD] 2025). Separately, a fast-growing commentary documents Bangladesh's exposure to imported-fuel shocks, most recently the 2026 Strait of Hormuz disruption (International Growth Centre [IGC] 2026; LightCastle Partners 2026).

What is largely missing is a comparative treatment that connects these strands: most work treats Pakistan's boom as a national success story, or Bangladesh's rooftop potential as a domestic policy question, but few extract systematic, governance-and-security-focused lessons from Pakistan for an industrially-anchored Bangladeshi transition. This paper addresses that gap by framing rooftop solar comparatively, as an energy-security response to import-fuel vulnerability, and by translating Pakistan's experience into a sequenced roadmap for Bangladesh.

3. Methodology and Research Design

3.1 Research design and justification

The study adopts a mixed-methods comparative case study, combining a qualitative cross-country comparison with an embedded quantitative techno-economic component. This follows from the research questions: **RQ1** and **RQ3** are explanatory and governance-oriented and are best addressed qualitatively, while **RQ2** and the roadmap are feasibility questions requiring quantitative grounding in capacity, solar-potential, and cost data. A purely qualitative design could not establish techno-economic viability; a purely quantitative one would miss the tariff, institutional and political-economy lessons central to the comparison.

3.2 Comparative case selection

Pakistan and Bangladesh are selected under a most-similar-systems logic. They share the key background conditions — neighbouring South Asian developing economies, heavy imported-fuel dependence, large export-oriented textile sectors, comparable solar irradiance and exposure to the same 2026 global fuel shock — while differing sharply on the variable of interest: the scale and speed of distributed-solar adoption. Holding the context roughly constant isolates the role of solar adoption in shaping energy-security outcomes, which is what makes Pakistan a meaningful natural experiment for Bangladesh, consistent with the logic of comparative case-study design (Yin 2018).

3.3 Data sources, methods, and verification

The study triangulates secondary data across four tiers: official statistics (NEPRA, BPDB, SREDA, IRENA, REN21, the World Bank); peer-reviewed analysis for the framework and techno-economic methods; industry and policy reports (IEEFA, TransitionZero, CPD, IDCOL); and credible journalism, used only for very recent 2025–2026 events and always cross-checked. The quantitative strand uses descriptive analysis of capacity and cost data, rooftop-potential estimation by segment, and a levelized-cost-of-electricity (LCOE) comparison; every quantitative claim was checked against at least two sources, and each capacity figure states its metric explicitly to avoid conflation. Only publicly available data were used, with no human subjects.

4. Conceptual Framework: Momentum versus Management

The paper organizes the comparison through a simple two-axis framework. The first axis, deployment momentum, captures how quickly distributed solar is being installed. The second, governance or management, captures the institutional readiness of the grid, tariff structure and regulatory regime to absorb that growth without destabilizing the power system or shifting costs unfairly. Concretely, momentum is read from installed and newly added distributed capacity relative to the size of the power system, while management is read from four institutional elements: cost-reflective tariff design, a stable net-metering and buyback regime, mandatory installation registration and monitoring and grid-integration provisions for distributed generation. A country scores high on management when these are established before deployment scales, and low when they lag behind it. Crossing the two produces four positions, shown in Figure 1. A fast-but-unmanaged transition delivers rapid capacity and even accidental energy security, but invites grid instability, cross-subsidy collapse, and policy reversals. The underlying dynamic is well established in the energy-economics literature: under volumetric

tariffs and net metering, self-generation erodes utility revenue while the network's fixed costs are recovered from a shrinking base, shifting costs onto non-adopters and, in the limiting case, threatening a self-reinforcing "utility death spiral" (Eid et al. 2014; Felder & Athawale 2014; Castaneda et al. 2017). A slow-but-managed position has the rules in place but little deployment. The fast-and-managed quadrant — rapid yet stable — is the target.

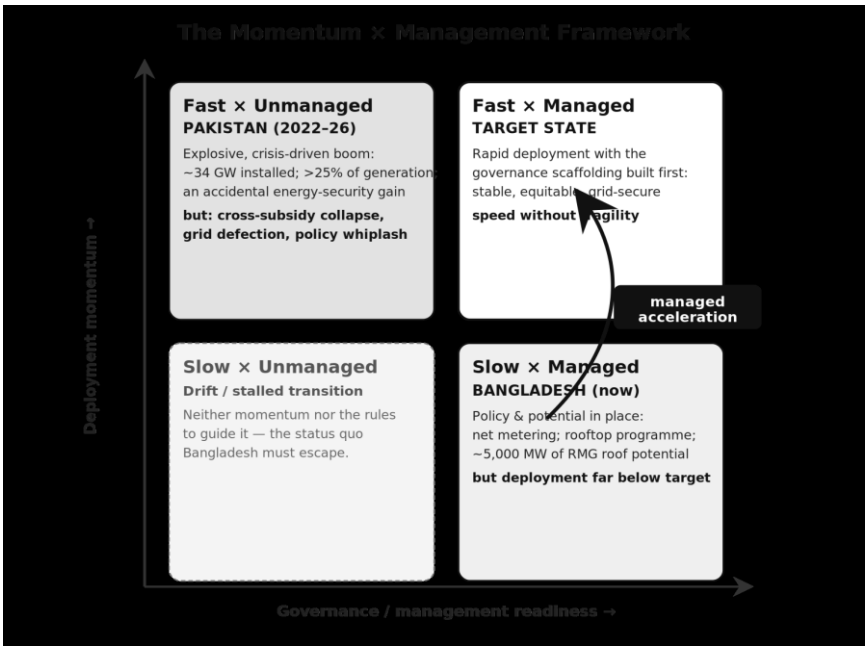


Figure 1. The momentum×management framework. Pakistan occupies the fast-but-fragile quadrant; Bangladesh sits in slow-but-managed; the argument is the diagonal move to fast-and-managed.

On this framework, Pakistan sits firmly in the fast-but-unmanaged quadrant: it achieved one of the world's fastest solar build-outs, but through a disorderly, consumer-led rush that its grid and tariff system were not designed to absorb. Bangladesh occupies the slow-but-managed quadrant: it has a net-metering policy, a national rooftop program, and abundant potential, yet minimal deployment. The paper's argument is that Bangladesh's task is to manufacture deliberately — through industrial policy and the energy-security imperative — the deployment momentum that crisis manufactured accidentally in Pakistan, while building the governance scaffolding before the rush rather than during it.

5. Two Energy Crises of Opposite Types

5.1 Pakistan: a grid-and-price crisis and an accidental solar boom

Pakistan's boom was driven from the bottom up by desperation, not by policy. As the power sector's circular debt mounted and annual capacity payments to idle plants alone exceeded PKR 2 trillion (about US$7 billion) in 2024, grid tariffs were raised steeply: average residential electricity prices roughly doubled between 2021 and 2024, with per-unit charges for many consumers reaching about PKR 44 by 2024, up from lifeline-slab rates of around PKR 9 a decade earlier (World Economic Forum 2025; Digital Pakistan 2026). At the same time, Chinese manufacturing overcapacity drove panel prices down sharply, from roughly US$0.35 per watt in the late 2010s to around US$0.10 per watt by the time the boom peaked in 2023–24 (TransitionZero 2025). Households facing steep marginal tariffs, farmers running diesel tube wells, and factories hedging against outages all turned to rooftop solar.

The scale is best captured by three metrics, because combining them understates the boom. Pakistan imported about 17 gigawatts (GW) of panels in 2024 alone — double the 2023 volume and the largest of any country that year and cumulatively close to 45–50 GW between 2019 and 2025, roughly the entire installed capacity of the national grid (World Resources Institute 2025; TransitionZero 2025). Net-metered rooftop capacity climbed from under 1 GW in the early 2020s to about 5.3 GW by April 2025, projected to pass 6 GW by year-end (REN21 2025). A bottom-up survey combining household data and satellite imagery estimated around 34 GW actually installed once off-grid and behind-the-meter systems are counted (TransitionZero 2025). By 2025, solar had become Pakistan's single largest electricity source, supplying more than a quarter of generation, up from about 14% the year before (World Economic Forum 2025). Figure 2 shows the three metrics together.

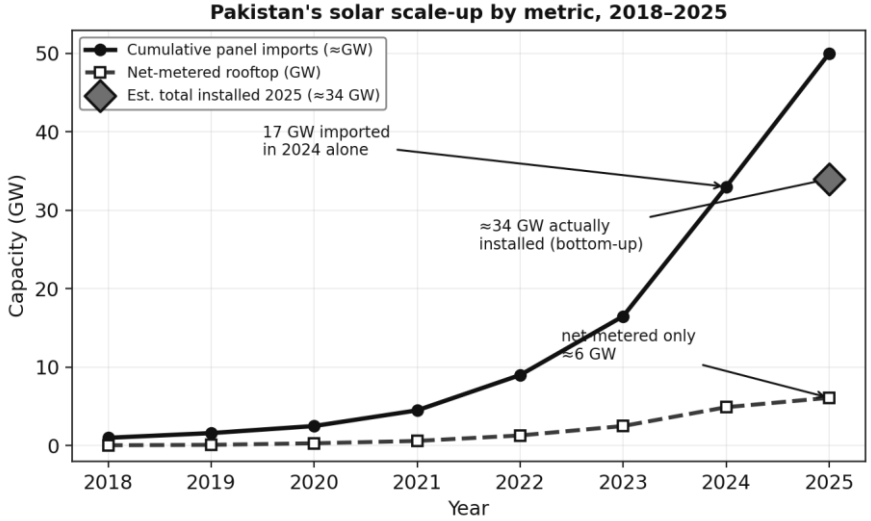


Figure 2. Pakistan's solar scale-up measured three ways. Reporting only net-metered capacity (≈6 GW) understates a boom that reached roughly 34 GW installed and 45–50 GW imported. Sources: World Resources Institute (2025); TransitionZero (2025); REN21 (2025).

Crucially, the boom delivered energy security as a by-product: agricultural electricity demand fell 34.3% in 2024 as tube wells went solar, and as self-generation spread the country cut its reliance on imported fuel and began renegotiating a long-term LNG contract with Qatar (World Resources Institute 2025; World Economic Forum 2025). When the 2026 fuel shock arrived, Pakistan's lower exposure to imported gas was, in part, a dividend of its disorderly solar rush.

5.2 Bangladesh: a fiscal-and-fuel paradox

Bangladesh's crisis is the structural inverse: the country does not lack power plants; it lacks the foreign exchange and fuel to run them. It holds substantial capacity yet relies heavily on imported LNG and petroleum, spending close to US$1 billion a year on more than six million tonnes of petroleum products, much of it routed through the Strait of Hormuz (IGC 2026). The binding constraint is fiscal and logistical, not the number of plants: when fuel markets convulse, the country cannot afford or secure enough fuel, and load shedding returns even as capacity sits idle. This is a chronic, structural condition rather than a one-off: Bangladesh experienced repeated fuel-driven load shedding across the 2022–2026 period as global gas prices spiked and dollar reserves tightened, and its exposure is deepening as domestic gas fields such as Bibiyana decline and the share of imported fuel in the power mix rises. The argument of this

paper therefore does not rest on any single geopolitical event; the structural vulnerability would remain even in a year without a crisis.

The 2026 Strait of Hormuz disruption is best read as the sharpest illustration of this standing vulnerability rather than its cause. After QatarEnergy declared unavoidable casualty in early March 2026, spot LNG prices Bangladesh had paid around US$10 per million British thermal units (MMBtu) in January surged to US$28.28 by mid-March (Global Voices 2026). With gas to power plants running at roughly 850–900 million cubic feet per day against demand of about 2,525, officials prepared for a 1,800 MW shortfall, ramped up coal, increased imports from India, and warned of load shedding (The Business Standard 2026a). Four of five state urea plants closed, freight and shipping costs for the ready-made garment (RMG) sector rose sharply, and the crisis was reported to have pushed as many as 1.2 million Bangladeshis into poverty (IGC 2026; LightCastle Partners 2026). Figure 3 summarises the shock.

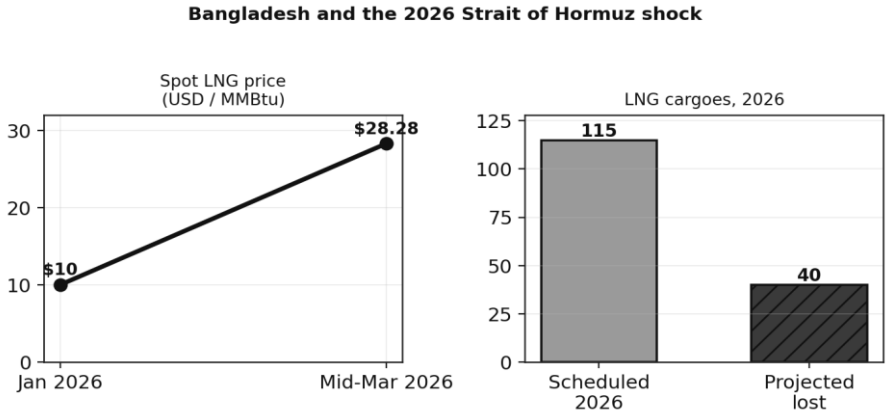


Figure 3. Bangladesh and the 2026 Strait of Hormuz shock: spot LNG prices nearly tripled within weeks, and roughly 40 of 115 scheduled LNG cargoes were projected lost. Sources: Global Voices (2026); The Business Standard (2026a).

This is the paradox at the heart of the comparison: Bangladesh suffers blackouts not from too few plants but from an inability to fuel the plants it has — an arithmetic of import dependence and thin reserves that recurs with or without a headline crisis. Pakistan's grid-and-price crisis pushed consumers to flee an expensive utility; Bangladesh's fiscal-and-fuel crisis leaves capacity stranded for want of affordable imported fuel. The causal sequence also matters: Pakistan's grid and public finances collapsed first, and that collapse caused the solar exodus; the energy-security payoff was accidental. Bangladesh should seek the same outcome deliberately, without first passing through grid collapse.

6. Comparative Diagnosis: Two Kinds of Resilience

Table 1 sets out the structural contrast that prevents a naive "copy Pakistan" reading. The two countries reached their crises through different mechanisms and must therefore pursue solar through different routes — Pakistan was pushed by tariffs, while Bangladesh must be pulled by industrial demand and energy-security need.

Table 1. Structural-conditions comparison

Dimension	Pakistan	Bangladesh

Primary push factor	Acute tariff crisis: prices ~doubled 2021–24; circular debt; unreliable grid	Import-fuel vulnerability, forex drain, recurring load shedding; pull from buyer compliance
Power-sector pathology	Shortage + high tariffs → grid defection	Idle capacity it cannot fuel → "capacity without security"
Exposure to 2026 fuel shock	Lower — distributed solar had cut grid and LNG demand	High — lost ~40 of 115 LNG cargoes; blackout risk; plant shutdowns
Lead adopters	Affluent households, agriculture (tube wells), industry	Export garment & textile factories (proposed anchor)
Policy posture	Reactive, whiplash (net metering → net billing, 2025–26)	Enabling on paper (Net Metering 2018/2025; rooftop programme 2025) but stalled
Central risk going forward	Cross-subsidy collapse, grid instability	Missing the energy-security and export-compliance window

Sources: compiled from World Resources Institute (2025); REN21 (2025); TransitionZero (2025); IEEFA (2025); Global Voices (2026).

7. Lessons to Replicate

Pakistan demonstrates two things Bangladesh should treat as proven. First, once rooftop economics beat grid power, adoption can scale with astonishing speed: a country can move from negligible distributed solar to tens of gigawatts within a few years driven by cheap modules and end-user economics rather than by central planning (World Resources Institute 2025; TransitionZero 2025). The latent demand exists wherever grid power is costly or unreliable — conditions that apply across Bangladesh's industrial base. Secondly, and most relevant to Bangladesh's specific crisis, distributed solar is an effective hedge against imported-fuel shocks. By shrinking grid demand and displacing gas-fired generation, Pakistan's solar build-out reduced its LNG appetite and softened its exposure to the very 2026 fuel shock that destabilised Bangladesh (World Economic Forum 2025). In the terms of the energy-security literature, which defines security as the continuity of energy supply relative to demand (Winzer 2012), distributed solar lowers the vulnerability of a vital energy system by simultaneously reducing its exposure to imported-fuel risk and strengthening its resilience (Cherp & Jewell 2014). For a country that loses dollars and power every time the Strait of Hormuz is threatened, this is the decisive lesson: rooftop solar is not merely a climate measure but an energy-security and balance-of-payments measure.

8. Lessons to Avoid

Pakistan's boom also offers a catalogue of governance failures that Bangladesh can prevent. The most serious is the regressive cross-subsidy shift, a mechanism formalized in the literature on net metering and network cost recovery (Eid et al. 2014; Brown & Sappington 2017). As wealthier consumers self-generated and bought less grid power, the fixed costs of the grid concentrated on those who remained — disproportionately poorer households. By December 2024, the government estimated that net-metered consumers had passed a burden of US$563 million (PKR 159 billion) on to other consumers, a figure forecast to rise to US$48.34 billion (PKR 4.24 trillion) by 2034 if the framework was not amended — the steep rise reflecting the compounding nature of the death-spiral feedback as each tariff rise prompts further defection rather than any single-year leap (REN21 2025). This fiscal pressure produced the second lesson: damaging policy whiplash. Rather than designing cost-fair tariffs from the outset, Pakistan responded late and abruptly. In 2025 the regulator moved to replace net metering with a less generous net-billing regime, proposing to cut the buyback rate for exported solar power from about PKR 27 per unit to roughly PKR 10–13 — a reduction of more than half that would extend payback periods and was paused only after public backlash

(Digital Pakistan 2026; REN21 2025). Such reversals damage investor confidence precisely when stable signals are needed. A third lesson is the data blind spot: because most installations occurred outside the formal net-metering system, regulators could not even measure true installed capacity, with off-grid and behind-the-meter systems estimated to dwarf registered capacity (TransitionZero 2025). Bangladesh can avoid each of these traps by designing cross-subsidy-proof tariffs, a predictable buyback glide-path, and mandatory installation registration before deployment accelerates.

Table 2. Transferable-lessons typology: what to replicate and what to avoid

Type	What Pakistan's experience shows	Implication for Bangladesh
Replicate	Once rooftop economics beat the grid, distributed solar scales to tens of GW within a few years.	Latent demand exists across the industrial base; deployment can move fast if enabled.
Replicate	Self-generation cut grid and LNG demand, softening exposure to the 2026 fuel shock.	Treat rooftop solar as energy-security and balance-of-payments policy, not a climate amenity.
Avoid	Regressive cross-subsidy shift: $563M (2024) onto non-solar consumers, projected $48.34B by 2034.	Design cost-fair, capacity-based tariffs from the outset.
Avoid	Policy whiplash: abrupt net-billing reversal (buyback PKR 27 → ~PKR 10–13).	Set a predictable, gradual buyback glide-path to protect investor confidence.
Avoid	Data blind spot: most installs sat outside net metering, so true capacity was unmeasurable.	Mandate registration and online monitoring of every installation from the start.

Source: authors' synthesis of Sections 7–8, drawing on World Resources Institute (2025), REN21 (2025), TransitionZero (2025), and World Economic Forum (2025).

9. The Bangladesh Pathway: Industrially-Anchored Managed Acceleration

Because Bangladesh cannot rely on a crisis-driven consumer rush, it must engineer demand deliberately, anchored where the alignment of incentives is strongest. The paper proposes a three-phase roadmap. Figure 4 shows why the garment sector is the natural anchor; Figure 5 sets out the phased timeline.

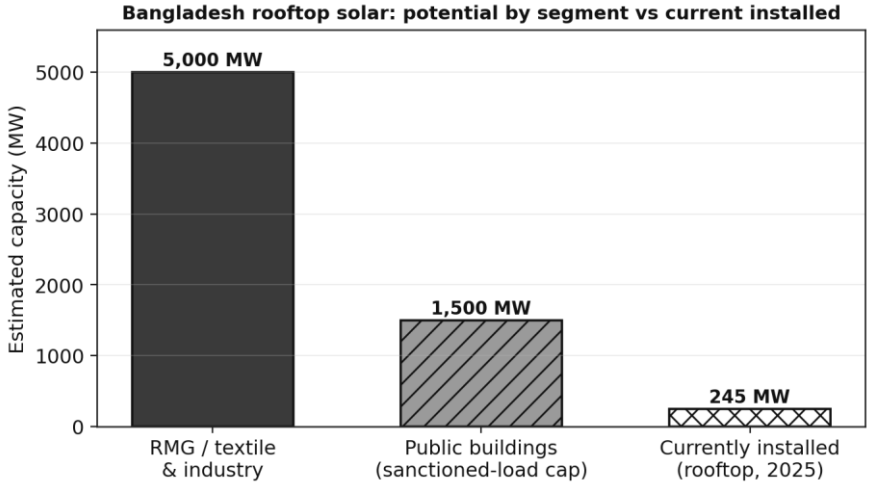


Figure 4. Bangladesh rooftop solar potential by segment versus current installed capacity. The garment/textile and industrial segment (≈5,000 MW, IDCOL estimate) dwarfs the public-building ceiling and today's installed base, making it the logical Phase 1 anchor. Sources: The Business Standard (2025); IEEFA (2025).

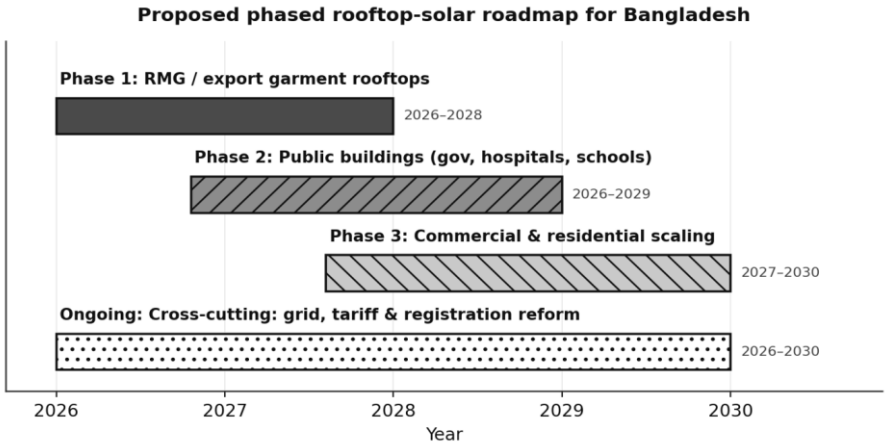


Figure 5. Proposed phased rooftop-solar roadmap, anchored on the export garment sector and underpinned by parallel grid, tariff, and registration reform.

Table 3. The three-phase deployment pathway at a glance

Phase	Target segment	Indicative potential	Financing model	Role and main limitation
Phase 1	Export garment & textile factories	~5,000 MW	Third-party OPEX	Anchor: scale, bankable offtakers, buyer-compliance pull.
Phase 2	Public buildings (offices, hospitals, schools)	<1,500 MW	CAPEX + OPEX	Accelerator: daytime load match; capacity-capped, cannot anchor.
Phase 3	Commercial & residential prosumers	Diffuse / large	Net metering	Scale-up: fragmented, low-load; needs feeder hosting-capacity limits.

Sources: The Business Standard (2025); IEEFA (2025); Government of Bangladesh (2025); pv magazine (2025).

9.1 Phase 1 — the export garment anchor

The ready-made garment sector is the natural anchor. Bangladesh has more than 7,000 RMG factories with large, contiguous rooftops, and the Infrastructure Development Company Limited (IDCOL) estimates that garment, textile, and other industrial rooftops could host around 5,000 MW (The Business Standard 2025). Peer-reviewed modelling supports the sector's viability: a simulated design for a Bangladeshi garment facility returned an LCOE of about US$0.091/kWh, comfortably below industrial grid and diesel-backup costs (Miskat et al. 2023). First movers already exist, demonstrating both feasibility and the third-party OPEX model, in which a developer installs and operates the system and the factory buys power below grid tariff (The Business Standard 2025). Five features make the garment sector the right place to begin, and to begin before the alternatives. First, scale of contiguous roof area: a single export factory offers more uninterrupted rooftop than dozens of scattered commercial or residential sites combined. Second, concentrated, daytime-coincident demand, so a large share of generation is consumed on site rather than exported. Third, creditworthy, bankable off-takers — established exporters with audited accounts which makes third-party financing viable in a way that fragmented residential demand does not. Fourth, an existing financing channel in IDCOL. Fifth, a distinctive export-competitiveness pull: international apparel brands increasingly require suppliers to demonstrate renewable-energy use and green-factory certification, a pressure that public and domestic-commercial users simply do not face. (The European Union's Carbon Border Adjustment Mechanism does not currently cover textiles, which fall outside its six covered sectors; the compliance pressure on garments comes instead

from brand commitments, due-diligence rules, and the prospect of future scheme expansion.) The alternatives are weaker as a starting point: public buildings are capacity-capped and cannot anchor a national target (Section 9.2), while commercial and residential rooftops are fragmented and individually low-load. The garment sector uniquely combines scale, bankability, and an external compliance driver — and, above all, factories that self-generate are insulated from the load shedding that fuel shocks inflict on industry. One structural complication should be designed for, however: many garment units operate in rented or multi-tenant industrial buildings, creating a split incentive in which the party that owns the roof is not the party that pays the electricity bill. Virtual or group net-metering arrangements, which allocate generation credits across multiple tenant accounts, can resolve this and should be enabled within the net-metering framework.

9.2 Phase 2 — public buildings as accelerators

The government's National Rooftop Solar Program, approved on 29 June 2025, targets 2,000–3,000 MW nationwide, with government offices on a publicly funded CAPEX model and hospitals and schools under third-party OPEX arrangements (Government of Bangladesh 2025). Public buildings offer two advantages beyond capacity: their daytime-peaking load matches solar output, maximizing self-consumption, and mandated installations de-risk the market and build installer capacity. A sector-wide peer-reviewed study illustrates the scale: rooftop systems on 663 public hospitals could supply roughly 90 MW and meet about 79% of that sub-sector's electricity demand, subject to facility-level feasibility assessment (Kowsar et al. 2025). Their limitation must be acknowledged, however: the combined sanctioned load of government offices, hospitals, and educational and religious institutions is less than 1,500 MW, so public buildings alone cannot meet the national target and function as accelerators rather than the anchor (IEEFA 2025).

9.3 Phase 3 — commercial and residential scaling

With an installer base and financing channels established, deployment can broaden to commercial buildings such as shopping malls — another daytime-peaking segment — and then to residential prosumers. The 2025 net-metering reforms support this widening: SREDA now allows net-metered systems up to 100% of sanctioned load, up from 70%, and admits single-phase consumers for the first time (pv magazine 2025). This phase, however, introduces a technical constraint that the industrial-anchor phases largely avoid: low-voltage distribution feeders in dense urban areas have a finite hosting capacity, and clustered residential injection can cause voltage rise and reverse power flow. Managing it requires smart inverters with volt-VAR control, feeder reinforcement where needed, and staged interconnection limits, so hosting-capacity assessment should precede mass residential rollout.

10. Engineering and Technical Considerations

Realizing this roadmap depends on sound engineering. Bangladesh's climate is favourable, with average irradiance of roughly 4.5–5.0 kWh per square meter per day, but high ambient temperatures impose penalties, as crystalline-silicon output declines roughly linearly as cell temperature rises above the standard test value. To project true DC yield under local thermal stress, designs should apply the standard temperature-coefficient derating model:

$$P_{dc} = P_{stc} \times [1 + \gamma \times (T_{cell} - T_{stc})] \quad (1)$$

where P_{dc} is the projected DC output, P_{stc} is the nominal module capacity under standard test conditions, γ is the manufacturer-specified temperature coefficient (typically about −0.3% to −0.4% per °C for crystalline silicon), T_{cell} is the actual operating cell temperature, and T_{stc} is the 25°C reference temperature. Realistic yield estimates must apply this derating rather than relying on nameplate capacity. Each industrial installation also requires a structural assessment of the existing roof, since many older factory sheds were not designed for the additional dead load and wind uplift of a PV array; a feasibility study covering structural capacity, shadow analysis, and tilt optimization should precede installation (CPD 2025).

Grid integration is the central system-level challenge. Bidirectional metering and standards-compliant inverters are mandatory under the net-metering policy, and at higher penetrations distribution utilities must manage voltage rise, power-factor correction, and frequency stability on local feeders (CPD 2025). The Pakistani experience shows that allowing large volumes of behind-the-meter generation without monitoring undermines both grid management and planning; Bangladesh should therefore require online inverter monitoring and registration from the outset. Coupling rooftop solar with battery storage – as Pakistan is increasingly doing having imported an estimated 1.25 GWh of lithium-ion batteries alongside panels in 2024 – can shift midday generation to evening peaks and ease grid integration, though at additional cost (World Resources Institute 2025).

11. Financing and Economic Viability

The economic case rests on a favourable levelized cost of electricity. With panel prices at historic lows and industrial tariffs rising, rooftop solar now undercuts grid and especially diesel-backup power for most industrial users, with paybacks well within the system's twenty-year life; Bangladeshi studies report LCOEs of US$0.05–0.09/kWh for well-sited systems (Miskat et al. 2023). The barrier is not viability but finance and policy friction. Two models are used: CAPEX, where the user funds the system, and OPEX, where a third party installs, operates, and sells power below grid tariff — removing the upfront cost that deters many factories (The Business Standard 2025). IDCOL is well placed to scale this, having pioneered one of the world's largest off-grid solar programmes by financing more than 4.1 million solar home systems serving roughly 20 million people (World Bank 2021), though that program's later decline as the grid expanded is a reminder that financing instruments must evolve with the market.

Two policy frictions stand out, the first of which Bangladesh has just decisively addressed. Until mid-2026, import duties on balance-of-system components were high and uneven — an effective burden of roughly 20% or more, falling unevenly across modules (about 26–37%), inverters, mounting structures, and DC cables — which inflated total project cost well above the module price and blunted the LCOE advantage (The Business Standard 2024). The FY2026–27 national budget, announced in June 2026, eliminated import, regulatory, and supplementary duties and advance tax on the full range of solar components — modules, inverters, mounting structures, DC cables, batteries, and storage systems — cutting the effective burden to zero, lowering the cost of a one-megawatt rooftop installation by an estimated 25–30%, and adding an income-tax holiday for solar generation to 2035 and a 5%

rebate on solar users' electricity bills (pv magazine 2026; The Business Standard 2026b). This is precisely the rationalization the comparison recommends; the priority now shifts from setting the duties right to ensuring quality standards keep pace, since zero duties without enforced equipment standards risk flooding the market with substandard imports (The Business Standard 2026b).

Table 4. Solar import-duty burden before and after the FY2026–27 budget

Component	Until mid-2026 (effective)	From FY2026–27
PV modules	~26–37%	0%
Inverters	high; uneven	0%
Mounting structures & DC cables	up to ~58%	0%
Batteries & storage systems	high	0%
Overall effective burden	~20% or more	~0% (plus income-tax holiday to 2035)

Sources: The Business Standard (2024 2026b); pv magazine (2026).

Second, beyond the firm-level payback lies a macro-financial return that national policy should price in: every megawatt of rooftop solar displaces imported LNG and diesel, easing the foreign-exchange and subsidy bleed that the 2026 shock made so costly. One limitation should be stated plainly: because solar generation is diurnal, rooftop PV on its own is only a partial hedge against the evening-peak load shedding that fuel shocks often trigger. Pairing systems with battery storage — whose falling costs are steadily improving its economics, as Pakistan's parallel battery imports show — extends the hedge into the evening but adds cost, so deployment planning should weigh the levelized cost of storage against the avoided cost of outages rather than treating solar capacity in isolation. Concessional and blended finance from development partners, alongside green bonds, can lower entry costs while this market matures.

12. Governance and Policy Recommendations

The comparison yields a clear governance agenda: capture Pakistan's deployment speed and energy-security gains while engineering out its failures in advance. Six measures follow. First, design cost-fair tariffs from the outset — capacity-based connection charges and time-of-use pricing — so that distributed solar does not shift fixed grid costs onto poorer non-solar consumers, the failure that has cost Pakistan dearly. Second, set a predictable, gradually-adjusting buyback glide-path rather than risking the abrupt net-billing reversals that damaged Pakistani investor confidence. Third, mandate registration and online monitoring of every installation, closing the data blind spot before it opens. Fourth, consolidate and safeguard the June 2026 duty reform: the FY2026–27 budget's removal of duties across all balance-of-system components (Section 11) has compressed levelized costs and released stalled private capital, but the gain must be protected by enforcing equipment-quality standards through SREDA and BSTI, so that zero duties accelerate good installations rather than flooding the market with substandard imports. This rationalization should be calibrated rather than indiscriminate: where the higher duties genuinely protect a nascent domestic assembly base for inverters or mounting structures, reform should preserve targeted, time-bound infant-industry support while removing duties on components that have no local manufacturing to protect. Fifth, build engineering-procurement-construction (EPC) capacity deliberately: with only an estimated 15–20 high-quality EPC firms operating nationally, the 3,000 MW target is unachievable on its current timeline without a workforce and quality-assurance plan (IEEFA

2025). Sixth, pair rooftop growth with investment in grid flexibility and storage so the system can absorb rising penetration. Throughout, distributed solar should be treated explicitly as energy-security and foreign-exchange policy, with the recurring fuel shocks and buyer-compliance deadlines used as the forcing mechanisms for action.

13. Discussion: SDG Alignment and Implications

The managed-acceleration pathway advances several SDGs at once: SDG 7 (affordable clean energy) through cleaner, more reliable industrial power; SDG 9 (industry and infrastructure) through resilient energy infrastructure and grid modernization; SDG 8 (decent work) by protecting the garment sector's competitiveness and jobs against load shedding and tightening buyer requirements; and SDG 13 (climate action) by cutting emissions and import dependence. The through-line is resilience: reframing rooftop solar as energy security aligns climate goals with the immediate imperative of insulating the export economy from fuel shocks.

14. Limitations and Future Work

Three limitations should be noted. First, the study relies on secondary data and does not include primary fieldwork or factory-level surveys, which would refine the rooftop-potential and LCOE estimates. Second, the empirical landscape is fast-moving: figures for the 2025–26 period, including the unfolding Strait of Hormuz disruption and Pakistan's evolving net-billing rules, remain provisional and should be updated at the point of submission. Third, distributed-solar capacity is genuinely difficult to measure, and the paper has therefore reported multiple metrics rather than a single figure. Future work should pursue primary techno-economic surveys of garment-factory rooftops, grid-integration modelling at higher penetrations, and quantitative evaluation of financing instruments for converting remittances and concessional funds into rooftop investment.

15. Conclusion

Bangladesh should follow the sun — but not Pakistan's path to it. The two countries' crises were of opposite types: Pakistan's grid-and-price collapse drove a disorderly consumer rush into solar that, almost by accident, bought it real energy security; Bangladesh's fiscal-and-fuel paradox leaves it with idle plants it cannot fuel and an export economy exposed to every shock through the Strait of Hormuz. Pakistan therefore serves as both model and warning. Its lesson to replicate is that distributed solar can rapidly cut imported-fuel dependence and strengthen energy security; its lesson to avoid is that an unmanaged boom invites cross-subsidy collapse, grid instability, and policy whiplash. The path forward for Bangladesh is managed, industrially-anchored acceleration — beginning with the export garment sector, scaling through public and commercial buildings, and underpinned from the start by cost-fair tariffs, registration, EPC capacity, and grid investment. Pursued deliberately, this captures Pakistan's gains while sidestepping its mistakes, and does so before the next fuel shock, and before tightening buyer mandates, make the choice for Bangladesh's exporters.

AI Use Declaration

In preparing this paper, the authors used artificial-intelligence tools for limited, clearly defined tasks. Figma and Google Gemini's image-generation model ("Nano Banana") were used to design and render the figures. Jenni AI was used to assist with language refinement, grammar and consistency checking, and reference formatting. All research design, data collection, analysis, interpretation, and conclusions are the authors' own; the authors reviewed and verified every AI-assisted output and take full responsibility for the final content of this paper.

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